

Hypothesis-Search Report — Rosacea /

phyma_fibroblast_identity

Hypothesis label: Rhinophyma fibroblasts are the PTGDS-high pro-inflammatory population of papulopustular disease, not a scar-type myofibroblast. **KB status:** EMERGING · **Applies to:** Phymatous Rosacea **Search/analysis date:** 2026-09-06 · **Iterations:** 1–4 of 5

1. Executive Judgment

Verdict: PARTIALLY SUPPORTED — the hypothesis is right in spirit but wrong on its named biomarker.

The seed hypothesis bundles two separable claims:

- **(A) Specific/biomarker claim:** rhinophyma fibroblasts carry the **PTGDS-high** pro-inflammatory signature of papulopustular rosacea.
- **(B) Conceptual/identity claim:** the phyma fibroblast is an **inflammation-driven state**, *not* a persistent ACTA2/COL1A1 scar-type myofibroblast, which explains rhinophyma's post-surgical reversibility.

A direct, first-hand computational test against the dataset named in the hypothesis (**ArrayExpress E-MTAB-16629**, actually downloaded and analysed) yields:

- **Claim (A) is CONTRADICTED in this atlas.** PTGDS is *lowest* in rhinophyma fibroblasts (5.0% positive) and *highest* in healthy-skin fibroblasts (54.5%), with hypertrophic scar intermediate (22.7%). PTGDS here behaves as a **homeostatic** dermal-fibroblast marker (co-tracking CFD/APOD), not an inflammatory disease marker, and the little PTGDS in rhinophyma localizes to **endothelium**, not fibroblasts. There is no hidden PTGDS-high fibroblast subpopulation.
- **Claim (B) is QUALIFIEDLY SUPPORTED.** Rhinophyma fibroblasts (i) show ACTA2 activation (38.9%) yet (ii) **lack the POSTN/CTHRC1 myofibroblast-persistence program** that defines hypertrophic scar (POSTN 3.1% vs 46.4%; CTHRC1 8.0% vs 43.4%), and (iii) express **reticular/inflammatory chemokines above scar levels** (CCL19 3.1×, CXCL12 1.9×,

CCL2 1.7× the scar fraction). So they are inflammatory-but-non-persistent — consistent with the hypothesis's "not a classical scar myofibroblast" and with reversibility.

Most important caveats: the referenced dataset has **one individual per condition** (pseudo-replication; no cross-donor statistics are possible), lineage was assigned by canonical-marker scoring rather than full integration, and — critically — the papulopustular reference (Chen et al.,

P 39384741

has no public accession, so the comparison used **published/proxy marker genes rather than a joint embedding**. PTGDS was therefore operationalized as a single gene plus a proxy inflammatory set, not Chen et al.'s exact pro-inflammatory-fibroblast signature. The verdict should be read as "the specific PTGDS-transfer claim fails in the only accessible dataset; the broader inflammatory-non-scar identity is plausible and partly supported, pending replication."

2. Evidence Matrix

#	Citation	Type	Stance	Mechanistic claim tested	Key finding	Subtype
1	P 39384741 (Chen 2024)	Human clinical + model organism	Supports (seed basis)	PTGDS+ pro-inflammatory fibroblasts are causal in rosacea	Papulopustular rosacea shows expansion of pro-inflammatory fibroblasts; fibroblast depletion or PTGDS knockdown blocks murine rosacea	Papulopustular (erythematous) inflammation
2	E-MTAB-16629 reanalysis (this report, computational)	Computational (scRNA-seq)	Refutes (A)	Do rhinophyma fibroblasts carry the PTGDS signature?	PTGDS+ fibroblasts: Healthy 54.5% > HS 22.7% > Rhinophyma 5.0% ; PTGDS is homeostatic here	Rhinophyma healthy
3	E-MTAB-16629 robustness (this report)	Computational	Refutes (A)	Is there a hidden PTGDS-high fibroblast subset?	PTGDS in only 2.9% of rhinophyma cells; PTGDS-high cells are endothelial ; stringent fibroblasts 4.0% PTGDS+/46.9% ACTA2+	Rhinophyma
4	E-MTAB-16629 reanalysis (this report)	Computational	Qualifies/ supports (B)	Scar-persistence program in phyma?	POSTN 3.1% vs 46.4% (HS); CTHRC1 8.0% vs 43.4% — rhinophyma lacks the scar-	Rhinophyma

#	Citation	Type	Stance	Mechanistic claim tested	Key finding	Subtype
					persistence program	
5	E-MTAB-16629 reanalysis (this report)	Computational	Qualifies/ supports (B)	Is phyma fibroblast inflammatory?	CCL19 3.14×, CXCL12 1.88×, CCL2 1.71× higher than scar fibroblasts	Rhinoph
6	P 25421393 (Crawford 2015)	In vitro	Supports (B) mechanism	POSTN drives myofibroblast persistence	Periostin induces proliferation/ α -SMA/contraction & sustains HS myofibroblasts; knockdown lowers α -SMA	Hypertro
7	P 20531985 (Zhou 2010)	In vitro/ human tissue	Supports (B)	POSTN marks scar not inflammation	Periostin over-expressed in keloid/HS but "largely absent from sites of inflammation"	Scar vs i dermat
8	P 31066603 (Maeda 2019)	In vitro	Context (B)	IL4/ IL13→POSTN→RhoA/ ROCK→TGF- β 1 loop	Th2 cytokines and TGF- β 1 drive a self-reinforcing periostin loop in abnormal scar	Keloid/H
9	P 40478772 (Fang 2025)	In vitro + model organism	Competing/ qualifies	Is PTGDS an inflammatory-not-scar discriminator?	PTGDS+ pro-inflammatory fibroblasts are themselves profibrotic ; AT56 (PTGDS inhibitor) reduces fibrosis in SSC	Systemic

#	Citation	Type	Stance	Mechanistic claim tested	Key finding	Subtype
10	P 24563692 (Two 2014)	Review (mechanism)	Competing (upstream)	KLK5→LL-37 innate axis drives rosacea	Elevated KLK5 → cathelicidin/LL-37 → inflammation, vasodilation, vascular proliferation	Rosacea
11	P 22076323 (Meyer-Hoffert 2011)	Review	Competing (upstream)	Epidermal proteases in rosacea	KLK5-derived LL-37 fragments are proinflammatory/angiogenic; keratinocyte-origin	Rosacea
12	P 42106045 (Wu 2026)	Model organism (mouse)	Competing (upstream)	LL-37 model, mast-cell/NF-κB	LL-37-induced rosacea-like dermatitis with mast-cell activation, Th1/Th17, NF-κB	Rosacea
13	P 41898724 (Fouani 2026)	Review	Competing (upstream/parallel)	TRPV4–mast cell neuroimmune signaling	Direct TRPV4-dependent mast-cell activation in LL-37-driven rosacea	Rosacea inflamm
14	P 42423091 (Bayar Muluk 2026)	Review (clinical)	Qualifies (B)	Is phyma reversible by anti-inflammatory Rx?	Medical Rx (doxycycline/isotretinoin) helps only early inflammatory Grade I ; fibrotic Grades II–IV need surgery	Rhinoph
15			Context	Phyma histology		Phymato

#	Citation	Type	Stance	Mechanistic claim tested	Key finding	Subtype
	P 24206029 23197212	Human clinical (case)			Rhinophyma = sebaceous hypertrophy + lymphoedema + fibrosis; phyma = end-stage rosacea	

3. Data and Tool Use

3.1 Datasets / sources

Source	Accession / URI	Repo	Version/ snapshot	Retrieved	Status	Re
Rhinophyma vs HS vs healthy scRNA- seq	E-MTAB-16629 (https://www.ebi.ac.uk/biostudies/studies/E-MTAB-16629)	ArrayExpress / EBI BioStudies	Released 2026-02-18	2026-09-06	Accessed & analysed (API HTTP 200; 9 processed 10x files + SDRF/IDF downloaded)	Di to ex na th
Papulopustular rosacea scRNA- seq (Chen et al.)	none found	—	—	2026-09-06	Could not be verified — no public accession in cached full text	W th co ab fo pr m co
PubMed	14 queries (see logs/pubmed_search_log.md)	NCBI	live	2026-09-06	Accessed; some queries 0-hit or rate-limited (logged)	Re

Exact E-MTAB-16629 subset used: **all three samples** — ROS (rhinophyma), HS-H (hypertrophic scar, chest, 22 y/o male, S6), Skin-H (healthy). Organism *Homo sapiens*; assay 10x 3' v3; comparison = fibroblast marker profiles across the three conditions. **One individual per condition (no biological replicates)**. File paths, byte sizes, and the resolved API response are logged in [openscientist_artifacts/logs/dataset_access_log.md](#).

Databases **not** queried (declared, not checked): GenCC, ClinGen, GWAS Catalog, ClinicalTrials.gov, GEO for an independent rhinophyma cohort. These are logged as **unverified** in Knowledge Gaps, not as evidence of absence.

3.2 Analyses (input → method → output)

ID	Outcome	Input	Method	Output
<code>fibroblast_marker_comparison</code>	succeeded	E-MTAB-16629 (3 samples)	scanpy 1.12.1: QC (min_genes=200, mito<20%) → CP10k+log1p → lineage by marker score-argmax → fibroblast subset → per-gene mean & %positive + composite scores; seed=0	res fibr res line
<code>rhinophyma_ptgds_robustness</code>	succeeded	E-MTAB-16629 (ROS)	clustering-free: rank cells by PTGDS; lineage profile of PTGDS-high cells; stringent fibroblast gate	res rhin
Leiden subclustering of ROS	failed	E-MTAB-16629 (ROS)	HVG→PCA→neighbors→leiden	none limit
<code>inflammatory_chemokine_comparison</code>	succeeded	E-MTAB-16629	fibroblast %positive chemokines, Rhino vs HS	res fibr (loca
Joint embedding with papulopustular data	skipped	—	not possible: no public Chen et al. accession	none

Environment: `openscientist_artifacts/environment.txt` (python 3.12; scanpy 1.12.1; anndata 0.12.6; scipy 1.17.1; numpy 2.3.5; pandas 3.0.3; sklearn 1.8.0; seed 0). Executor constraints: 60 s wall-clock per call, no cross-call filesystem persistence, network via `requests` only — these forced per-sample processing and precluded full multi-sample integration and cell-level cross-condition statistics.

4. Mechanistic Causal Chain

Seed-implied chain (phyma arm): Upstream rosacea trigger → keratinocyte/innate activation → pro-inflammatory fibroblast expansion (**PTGDS+**, per papulopustular data) → sustained inflammatory + vasodilatory signalling → *phyma* as a persisting inflammatory-fibroblast state (not scar) → surgically reversible.

What the evidence supports vs infers:

- **Upstream trigger → innate/keratinocyte activation:** STRONG (KLK5/LL-37/TLR2/mast-cell/TRPV4; PMIDs 24563692, 22076323, 42106045, 41898724). This is keratinocyte/immune-centred, **upstream of and largely independent from** the fibroblast-identity question.
- **Papulopustular pro-inflammatory fibroblast is causal & PTGDS-marked:** STRONG for papulopustular
([P 39384741](#)),
but **stage- and subtype-specific**.
- **Same PTGDS+ state persists in phyma:** NOT SUPPORTED in E-MTAB-16629 (rows 2–3). This is the **broken link** in the seed chain.
- **Phyma ≠scar myofibroblast:** SUPPORTED via a different molecular route — phyma is ACTA2+ but **POSTN/CTHRC1-negative** (rows 4, 6–7), so it lacks the self-reinforcing POSTN→RhoA/ROCK→TGF-β1 persistence loop
([P 31066603](#)).
- **Reversibility:** best explained by **absence of the POSTN persistence program** plus surgical removal of sebaceous/fibroplastic bulk
([P 42423091](#)),
rather than by "inflammation that resolves." **Inferred**, not directly demonstrated longitudinally.

Missing causal steps: (i) a direct phyma↔papulopustular fibroblast lineage/state comparison in one embedding; (ii) demonstration that phyma fibroblasts are *derived from* the papulopustular pro-inflammatory state (no trajectory/longitudinal data); (iii) any perturbation showing PTGDS is functionally relevant *in phyma* (as opposed to papulopustular).

5. Knowledge Gaps

1. **PTGDS identity of phyma fibroblasts (conflicting evidence).** Scope: core claim (A).
Checked: E-MTAB-16629 (accessed) → PTGDS-low in rhinophyma fibroblasts; Chen et al. → PTGDS-high in papulopustular (no accession to re-test). Why it matters: the seed's biomarker fails to transfer in the only accessible atlas. Resolve: obtain/authorize the Chen papulopustular scRNA-seq and co-embed with E-MTAB-16629; add ≥ 3 –5 rhinophyma donors; IHC/RNAscope for PTGDS in phyma fibroblasts vs endothelium.
 2. **Single-donor pseudo-replication.** Scope: all computational claims. Checked: SDRF confirms $n=1$ /condition. Why: no cross-donor statistics; batch/depth confounds. Resolve: multi-donor cohort with mixed-effects modelling.
 3. **Proxy vs exact signature.** Scope: claims (A)/(B). Checked: no public Chen gene list. Why: PTGDS-only + proxy chemokines may misrepresent the true papulopustular signature. Resolve: apply the published exact signature once available.
 4. **Ambient-RNA confound for chemokines.** Scope: claim 5. Checked: rhinophyma sample is immune/endothelial-rich. Why: fibroblast chemokine fractions may be inflated by spillover. Resolve: SoupX/CellBender decontamination; spatial validation.
 5. **Reversibility is inferred, not observed.** Scope: claim (B) clinical arm. Checked:

P 42423091

 (staging), but no longitudinal omics before/after debulking. Resolve: paired pre/post-surgery biopsies with recurrence follow-up.
 6. **Source/database absences (unverified, not "absent").** Scope: causal-graph completeness. Checked so far: PubMed only. **Not** queried: GenCC, ClinGen, GWAS Catalog, ClinicalTrials.gov, an independent GEO rhinophyma dataset. Resolve: run those queries with recorded version/date/filters before asserting absence.
 7. **PTGDS directionality is context-dependent.** Scope: mechanism. Checked:

P 40478772

 shows PTGDS+ fibroblasts are profibrotic in SSc. Why: PTGDS cannot cleanly separate "inflammatory-reversible" from "fibrotic-persistent." Resolve: functional PTGDS perturbation in phyma-derived fibroblasts.
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6. Alternative Models

Model	Relation to seed	Summary	Evidence
KLK5/ cathelicidin(LL-37)/TLR2 innate-immune axis	Upstream cause (keratinocyte-centred)	Serine-protease-driven LL-37 generation initiates inflammation/vasodilation/angiogenesis; fibroblast changes are downstream	P 24563692 P 22076323 P 23871720
Mast cell / TRPV4 neurogenic inflammation	Parallel/ upstream	LL-37→mast-cell degranulation & TRPV4 neuroimmune signalling drive erythema/edema	P 42106045 P 41898724
Demodex / microbiome persistence	Upstream cofactor	Follicular Demodex sustains innate activation; ivermectin efficacy	P 42544818 P 42541637
POSTN-negative activated-fibroblast model (this report)	Refinement of seed claim B	Phyma fibroblast = ACTA2+, chemokine-high, POSTN/CTHRC1-low → inflammatory, non-persistent, surgically reversible; PTGDS is <i>not</i> its marker	Rows 4–7
Sebaceous-hyperplasia + lymphoedema/ fibroplasia (structural)	Downstream/ parallel	Late phyma bulk is sebaceous + fibrotic tissue requiring surgery, not anti-inflammatory reversal	P 42423091 P 24206029
PTGDS+ profibrotic-fibroblast model	Competes with seed's dichotomy	PTGDS+ pro-inflammatory fibroblasts can <i>drive</i> fibrosis (SSc), so PTGDS ≠ "inflammatory not scar"	P 40478772

7. Discriminating Tests

1. **Joint scRNA-seq embedding, phyma vs papulopustular vs HS vs healthy** (≥ 5 donors each). Stratify by rosacea subtype/phyma grade. *Expected if seed true*: phyma fibroblasts co-cluster with papulopustular PTGDS+ fibroblasts. *Expected from this report*: phyma fibroblasts form a PTGDS-low, POSTN-low, chemokine-high cluster distinct from both papulopustular and scar.
2. **Spatial/IHC of PTGDS in phyma** (RNAscope PTGDS + DCN/PDGFRA fibroblast vs PECAM1 endothelial). *Expected from this report*: PTGDS predominantly endothelial, not fibroblast.
3. **POSTN/CTHRC1 vs chemokine staining across phyma grades I–IV**. *Expected*: POSTN stays low across grades (reversibility correlate); chemokines high in early/inflammatory phyma.
4. **Paired pre/post-debulking biopsies with 12-month recurrence follow-up + bulk RNA-seq**. *Expected*: low baseline POSTN predicts non-recurrence; contrast with HS.
5. **Functional PTGDS perturbation (AT56 or knockdown) in phyma-derived fibroblasts**. *Expected if seed true*: PTGDS inhibition reduces inflammatory output of phyma fibroblasts (as in papulopustular mice / SSc). If phyma fibroblasts are PTGDS-low, minimal effect — discriminating.

8. Curation Leads (require curator verification)

- **Status**: consider **EMERGING** → **CONTESTED/PARTIALLY-SUPPORTED** for claim (A) (PTGDS-high phyma fibroblast) based on the E-MTAB-16629 reanalysis; retain **EMERGING** for claim (B) (non-scar-myofibroblast identity).

- **Subtype restriction**: the PTGDS+ pro-inflammatory fibroblast is evidenced for **papulopustular** rosacea

([P 39384741](#));

do **not** extend the PTGDS marker to phymatous without direct phyma evidence.

- **Candidate nodes/edges**:

- Rhinophyma fibroblast —is_a→ activated fibroblast (ACTA2+, POSTN–/CTHRC1–) (computational lead).
- PTGDS —marker_of→ homeostatic/quiescent dermal fibroblast **and** dermal endothelial cell in phyma (computational lead; contradicts PTGDS→phyma-fibroblast edge).

- POSTN/CTHRC1 persistence loop —distinguishes→ hypertrophic scar from rhinophyma
(

P 25421393

P 20531985

P 31066603

).
- KLK5/LL-37 axis —upstream_of→ rosacea inflammation
(

P 24563692

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- **Candidate ontology terms:** fibroblast (CL:0000057), myofibroblast (CL:0000186), dermal fibroblast (CL:0002620), endothelial cell (CL:0000115); processes: chronic inflammatory response (GO:0002544), extracellular matrix organization (GO:0030198), myofibroblast differentiation (GO:0036446).
- **Candidate evidence snippets to verify:**
 - P 25421393

 :
"Hypertrophic scarring is characterized by the excessive development and persistence of myofibroblasts."
 - P 20531985

 :
"over-expressed in keloid and hypertrophic scars, atopic dermatitis, but is largely absent from sites of inflammation..."
 - P 40478772

 :
"Prostaglandin D2 synthase (PTGDS) ... enriched expression in pro-inflammatory fibroblasts associated with skin fibrosis."
 - P 42423091

 :
"Medical therapy, including doxycycline and isotretinoin, is primarily beneficial in early inflammatory disease (Clark Grade I). Established fibrotic rhinophyma (Grades II-IV) generally requires surgical intervention."
- **Candidate knowledge_gaps prompts:** add gaps #1 (PTGDS non-transfer conflict), #5 (reversibility not longitudinally shown), #6 (GenCC/ClinGen/GWAS/trials unverified), #7 (PTGDS context-dependent directionality).

9. Limitations & Future Directions

Primary limitation is the **single donor per condition** in the only accessible dataset and the **absence of a public papulopustular scRNA-seq accession**, which together prevent a definitive joint test and any cross-donor statistics. All computational claims are **exploratory and descriptive**. The strongest next step is a **multi-donor joint embedding of phyma, papulopustular, scar, and healthy skin**, paired with **PTGDS spatial validation** and **pre/post-surgery longitudinal sampling**. The report never silently substituted literature for the failed/limited analyses: the Leiden step is reported as failed, the joint embedding as skipped (no accession), and un-queried databases as unverified.

Artifact

bundle:

`kb/hypotheses/Rosacea/phyma_fibroblast_identity/openscientist_artifacts/`

(MANIFEST.yaml, code, derived tables, robustness note, search & access logs; figure noted local-only).